Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

JAVA ASSIGNMENT

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SAP ID:60017230102

ROLL NO.:A018

BRANCH:AIML

BATCH:A1

DATE OF SUBMISSION:5/04/24

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Assignment - 6

Aim/Objective :-
- To implements array of objects and passing/returning objects.

⇒ Syntax for creating multiple objects and arrays of objects
• Creating multiple objects of class Hello

```
Hello obj1 = new Hello();
Hello obj2 = new Hello();
```

General :-

```
<class name> <obj name> = new <class name>();
```

• Creating an array of object :-

```
<class name> [] <array name> = new <class name> [<size>];
```

↳ for declaring array

⇒ for assigning values (assuming constructor exists)
↳

```
for (int i = 0; i < <size>; i++) {
  <array name> [i] = new <class name> (param 1, param 2 ....);
}
```

Pass by ~~value~~ Pass by ~~reference~~ :-

Pass by value
Pass by value means passing a parameter to ~~the~~ m the method as value. It ~~does~~ not affect the original parameter i.e. change in the passed parameter isn't reflected

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in the original parameters

Syntax :-

<obj name>. <method-name> (param 1, param 2....); ← actual parameters

Pass by reference :-
In pass by reference, a reference of the object is passed to the method instead of a parameter. A change in the parameters of the formal object is reflected in the actual object.

Syntax :-

<obj name>. <methodname> (ref 1, ref 2...);

This pointer in java
The 'this' keyword is used to refer to current class instance variable.
It solves the issue of ambiguity incase of variables having same name as instance variables

Syntax :-

this. <instance variable name>

Website references :-
java tpoint.org

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#WAP to perform mathematical operations on 2 complex no. by passing and returning object as argument.

```
import java.util.*;
class Complex {
double real, imaginary;
Complex (double real, double imaginary) {
this.real = real;
this.imaginary = imaginary; }

Complex add (complex s) {
double r = this.real + s.real;
double i = this.imaginary + s.imaginary;
return new Complex (r, i); }

Complex subtract (complex s) {
double r = this.real - s.real;
double i = this.imaginary - s.imaginary;
return new Complex (r, i); }

class main { public static void main (String args[]) {
double i, r;
System.out.println ("Enter first complex no. a+ib:");
Scanner sc = new Scanner (System.in);
r = sc.nextDouble();
i = sc.nextDouble();
Complex num 1 = new Complex (r, i);
```

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```
System.out.println("Enter the second complex number x+iy:");
r=sc.nextDouble();
i=sc.nextDouble();
Complex num2 = new Complex(r,i);

Complex sum = num1.add(num2);

Complex d = num1.substract(num2);
System.Out.print ln ("Entered numbers: \n Num1 = " + num1.real
        + "+i" + num.imaginary + "\n Num 2 = "
        + "+i" + sum.imaginary + "\n Difference = "
        + d.real + "+i" + d.imaginary);
}
}
```

Output:-

Enter the first complex number a+ib:
3
4
Enter the Second complex no. x+iy:
2
3
Entered numbers:
Num1 = 3 +i4
Num2 = 2+i3
Sum = 5+i7
Difference = 1+i1

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Experiment No: 6

Q1] WOOP to arrange the names of students in descending order of their total marks, input data consists of students details such as names, ID.no, marks of maths, physics, chemistry. (Use array of objects)

CODE:

```
import java.util.Scanner;
class student {
 int IdNo,phy,chem,maths,n,total;
 String name;
        student(int id, String n,int m,int p, int c, int t) {
                IdNo=id;
                 name=n;
                 maths=m;
                phy=p;
                 chem=c;
                total=t;
        }
        public static void main(String[] args) {
                int IdNo,phy,chem,maths,n,total;
                 String name; student s[];
```


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```
        Scanner sc= new Scanner(System.in);
        System.out.println("Enter the number of students : ");
         n=sc.nextInt();
         s=new student[n];
        System.out.println("Enter records of "+n+" student : ");
        for(int i=0;i<s.length;i++)
        {
                System.out.println("Enter record of "+(i+1)+"
                student : ");
        System.out.println("Enter Id number of student : ");
        IdNo=sc.nextInt();
        System.out.println("Enter name of student : ");
        name=sc.next();
        System.out.println("Enter Marks
        of(Maths,Physics,Chemistry) : ");
        maths=sc.nextInt();
        phy=sc.nextInt();
        chem=sc.nextInt();
        total=phy+chem+maths;
        s[i]=new student(IdNo,name,maths,phy,chem,total);
    }
    System.out.println("Record of students as per decending
    order of total marks : ");
    for(int i=0;i<s.length-1;i++)
    {
```


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```
            for(int j=0;j<s.length-1-i;j++)
            {
                if(s[j].total<s[j+1].total)
                {
                    student temp;
                    temp=s[j];
                    s[j]=s[j+1];
                    s[j+1]=temp;
                }
            }
        }
            System.out.println("ID \t Name \t Maths\t
            Physics\tChemistry\tTotalMarks ");
            for(int i=0;i<s.length;i++)
            {
            System.out.println(s[i].IdNo +"\t"+s[i].name +"\t
            "+s[i].maths +"\t "+s[i].phy +"\t\t
            "+s[i].chem +"\t\t"+s[i].total);
            System.out.println();
            }
        }
}
```

OUTPUT:

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```
Enter the number of students :
2
Enter records of 2 student :
Enter record of 1 student :
Enter Id number of student :
1122
Enter name of student :
Naina
Enter Marks of (Maths,Physics,Chemistry) :
93
88
97
Enter record of 2 student :
Enter Id number of student :
1127
Enter name of student :
Aditi
Enter Marks of (Maths,Physics,Chemistry) :
95
85
97
Record of students as per decending order of total marks :
ID        Name     Maths    Physics        Chemistry        TotalMarks
1122     Naina      93        88              97              278

1127     Aditi      95        85              97              277
```


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Q2] WAP to perform mathematical operations on 2 complex numbers by passing and returning object as argument. Show the use of this pointer.

CODE:

```
public class Complex
{ double real;
double imag;
public Complex(double real, double imag)
{
this.real = real;
this.imag = imag;
}
public static void main(String[] args)
{
Complex n1 = new Complex(2.3, 4.5),
n2 = new Complex(3.4, 5.0),
temp;
temp = add(n1, n2);
System.out.printf("Sum = %.1f + %.1fi", temp.real, temp.imag);
}
public static Complex add(Complex n1, Complex n2)
{
Complex temp = new Complex(0.0, 0.0);
temp.real = n1.real + n2.real;
```


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```
temp.imag = n1.imag + n2.imag;
return(temp);
}
}
```

OUTPUT:

```
Microsoft Windows [Version 10.0.19045.4170]
(c) Microsoft Corporation. All rights reserved.

C:\Users\lenovo>cd C:\java program

C:\java program>javac Complex.java

C:\java program>java Complex
Sum = 5.7 + 9.5i
C:\java program>
```


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CONCLUSION:

• We learnt the implementation of an array of objects and the use to pass by value
and pass by
reference method
• We also understood how to create objects that act like pointers in Java
• We also saw the use of 'this' reference

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